

MAE 223- Ocean Technology Devices

MissionCreep11Fans Buoy Analysis

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1. Table of Contents

1. Table of Contents.....	1
2. Introduction:.....	2
3. Design and Development:.....	2
3.1. Electrical System.....	2
3.2. Mechanical System.....	3
3.3. Drogue.....	4
4. Testing and Results:.....	5
4.1 TEST 1 : RTK Status Data Acquisition - Campus Adventure.....	5
4.1.1. Pendulum Test - Hardware Troubleshooting.....	5
4.1.2. Alternative Test - Campus Adventure Test Parameters.....	5
4.1.3. Campus Adventure Results.....	5
4.2 TEST 2: Ferris Wheel.....	7
4.2.1. Ferris Wheel Test Parameters.....	7
4.2.2. Ferris Wheel Test Results.....	7
4.3 TEST 3: Salt water Pool.....	10
4.3.1. Salt water Pool Test Parameters.....	10
4.3.2. Salt water Pool Test Results.....	11
4.4 TEST 4: Ocean Deployment.....	11
4.4.1. Ocean Deployment Test Parameters.....	11
4.4.2. Ocean Deployment Test Results.....	11
5. Discussion/Limitation:.....	12
6. Conclusion:.....	12
7. Acknowledgment:.....	12
8. List of Materials and Budget.....	12
9. References.....	12

2. Introduction:

To assist in coastal monitoring, oceanographic research, and environmental modeling, it is necessary to understand tides and wave motions. Currently, existing marine instruments are expensive, difficult to deploy, and limited in portability and accessibility. The team aims to create a portable, low-cost marine device that could measure wave and tide motion using Global Positioning System(GPS) and Inertial Measurement Unit (IMU) data. This device shall achieve the following function requirements (FR):

#	Requirement
FR1	Precise and accurate positions and motion measurements of the ocean surface elevation to at least centimeter-level position
FR2	Budget-friendly
FR3	Portable & deployable
FR4	Collect and log data for long durations of time

3. Design and Development:

3.1. Electrical System

To achieve centimeter-level positional accuracy, the buoy integrated three main electrical components:

- SparkFun ZED-F9P Receiver: Real-Time Kinematic(RTK) Global Navigation Satellite System (GNSS) receiver
- Arduino ESP32 Thing Plus: provides wireless connectivity for receiving RTK correction data
- OpenLog Artemis (OLA): logs both GPS and IMU data onto an SD card

The SparkFun ZED-F9P Receiver can achieve centimeter-level positional accuracy when supplied with RTK correction data. It collects satellite positioning data while the ESP32 provides network connectivity. The team used Network Transport of RTCM via Internet Protocol (NTRIP) to deliver correction streams over the internet, enhancing accuracy. U-Center is used to configure and verify the GNSS receiver's settings and visualize the data output. The OLA records the corrected GNSS data from the ZED-F9P and logs the IMU measurements to an embedded Secure Digital (SD) card.

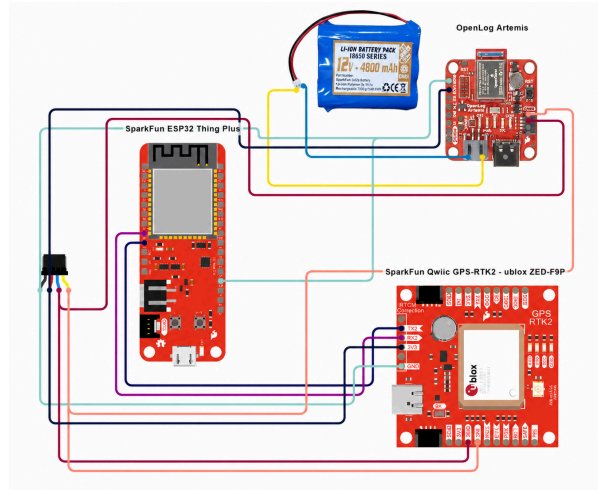


Figure 1: Wiring Diagram of the System

Figure 1 shows the components wired together through a custom diagram and mounted inside the enclosure onto an acrylic plate using standoffs. The components are powered by a lithium battery, while a phone provides a hotspot that is in the compartment underneath the buoy.

3.2. Mechanical System

The buoy enclosure comprises a spherical shell split into two halves, sealed with a locking clamp and an O-ring. Internally, a custom acrylic plate was designed to match the buoy's 8-hole mountain pattern, providing a stable surface for assembling the electrical components.

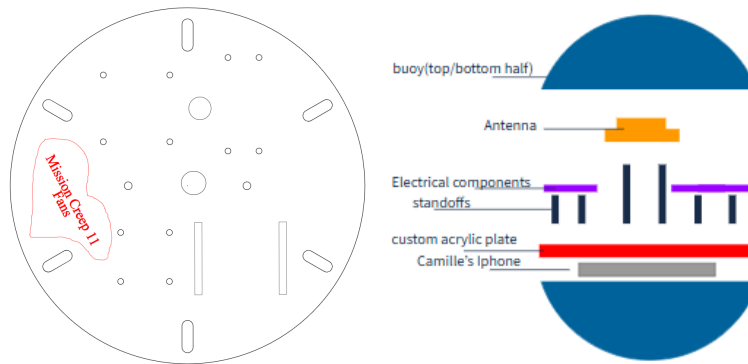


Figure 2: (a, left) Pictured is the custom acrylic plate that was cut with slotted holes and positioned holes to match the 8 holes that encompass the inside of the buoy and the holes for the standoffs to be placed. (b, right) a high-level figure of the assembly in an exploded view diagram

Since the 8-hole pattern had a threaded mate, the slotted holes widened the assembly tolerance required for the buoy. As shown in Figure 2b, the standoffs attach the components to the acrylic plate. The antenna standoffs are longer because a higher antenna position improves signal strength.

3.3. Droque

The team aims for the buoy to measure tidal motion, so, ideally, reducing the various forces that pull the buoy is advantageous and is achieved with a drogue. A drogue is a drag

device that dampens energy from waves and provides additional stability to a buoy. This would reduce the effects of wind on the buoy and improve sensor quality. The team settled on a vertical cone drogue design.



Figure 3: The physical drogue that is created with budget items

The drogue was constructed from budget materials, including a polypropylene bag, a kite, an aluminum ring, a custom acrylic hoop, zip ties, and paracord rope (Figure 3). The Acrylic ring was laser cut on an Epilogue Laser Engraver at the Design and Innovation Building Makerspace, and the aluminum ring was also machined there.

4. Testing and Results:

4.1 TEST 1: Campus Adventure

4.1.1. Pendulum Test - Hardware Troubleshooting

During pendulum testing week, we started having trouble with our system. Opening the files on the SD card, we noticed that IMU files were written, however, there weren't any u-blox files. Viewing the OLA serial monitor, we received the following errors: "GNSS Offline," "GNSS Not Detected," and "IMU initialized but data not ready." Assuming that the issue was between the OLA or the ZED boards, we began troubleshooting. We started by trying to isolate the OLA and ZED boards with known working systems with the help of our peers. Turning to the u-center interface, the data transmission (Tx) and receiving (Rx) readings were verified to work on our ZED board, confirming that our OLA board was faulty, requiring replacement.

4.1.2. Alternative Test - Campus Adventure Test Parameters

Our troubleshooting efforts resulted in an inability to test our system on the pendulum. The pendulum test's main goal was to gain an idea of the accuracy of the system after a duration of time, or, how well it maintained RTK Fixed status. With input from Professor Lucas, we set up a different test to ensure functionality of the system, and decided that this same goal could be achieved through simply going on a long walk around campus.

At this time, our buoy's acrylic plate was not yet made, so our boards were placed inside Markus's backpack with only the antenna pulled outside. The path we took was recorded, shown in Figure 4.

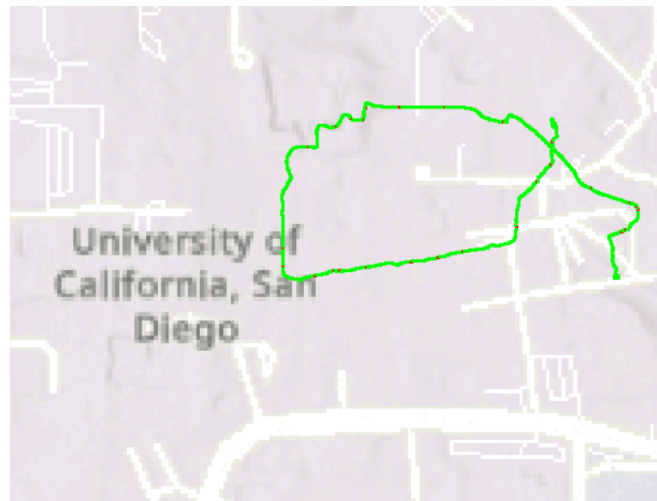


Figure 4: Raw u-blox file opened in u-center, plotting our path around campus in green.

4.1.3. Campus Adventure Results

Processing our files using the u-blox file parser, only 18.1% of data held RTK Fixed, with 62.4% in RTK Float, and 19.5% in 3D.

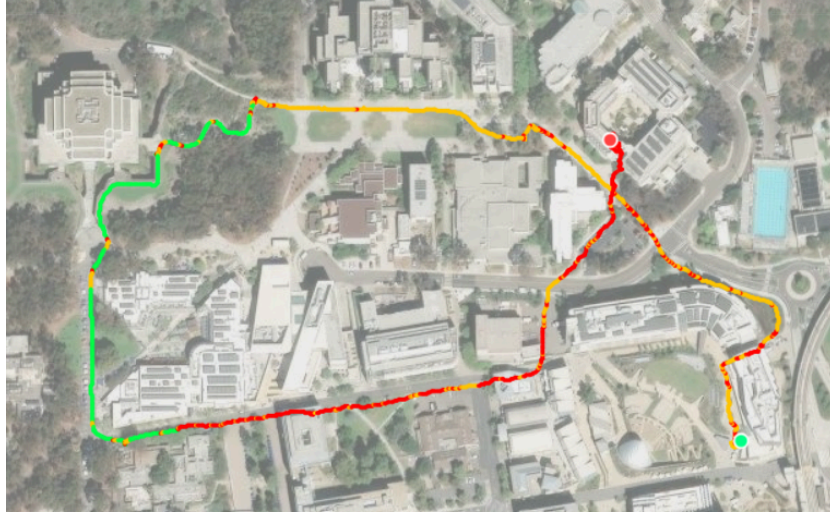


Figure 5: Parsed data plotted over a satellite view of campus, showing the path taken starting at the green dot and ending at the red dot. The green path shows RTK Fixed, orange RTK Float, and red recorded 3D/none.

Upon reflection of our setup and RTK status, we recalled that at the bottom of the snake path under Geisel library, we decided to hold the antenna at a higher position and upright, as opposed to its previous location placed on the outside of Markus's backpack. This corresponds to when we started getting RTK Fixed as shown in Figure 5. Then, when we lost RTK Fixed status, we recalled changing the antenna's position again. We realized the importance of keeping this antenna at this upright position with clear views of the sky.

From the parsed data, we also noted that although we had low RTK Fixed status, the 2D overhead view of our path is largely captured in Figure 5's plot. However, plotting in 3D, shown in Figure 6, the difference in accuracy between just Fixed and Float is significant, with even less desirable results with no RTK status.

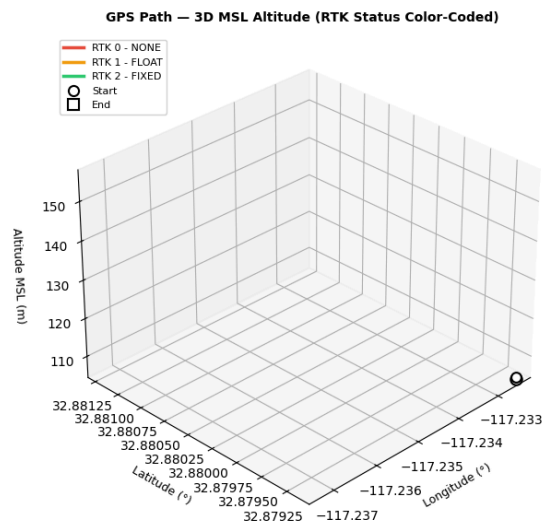


Figure 6: 3D plot of our GPS path, color-coded by RTK Status (Green - Fixed ; Yellow - Float ; Red - None)

4.2 TEST 2: Ferris Wheel



Figure 7 : Ferris Wheel Test Fixture

4.2.1. Ferris Wheel Test Parameters

Due to our inability to perform the pendulum test, we especially wanted data to achieve two goals during this second ferris wheel test: First, determine how effectively our system achieves RTK Fixed - and Second, analyze the ability of our system to capture the swing's periodic motion, as the ocean waves we will ultimately measure follow a similar circular pattern of movement.

For the ferris wheel test setup, the buoy is placed on a platform that swings in full circular motions at three different speeds in five minute intervals. Phase 1 was of moderate speed, phase 2 was fast, and phase 3 was slow. The platform remained horizontal for the duration of the test, keeping the buoy in an upright position.

4.2.2. Ferris Wheel Test Results

From the GPS data, we collected 63.0% RTK Float and 35.2% RTK Fixed data. From the animated plot of the buoy's position data in Figure 8, the rotational movement was captured. However, erratic behavior can be seen, largely due to the loss of RTK Fixed in the second phase.

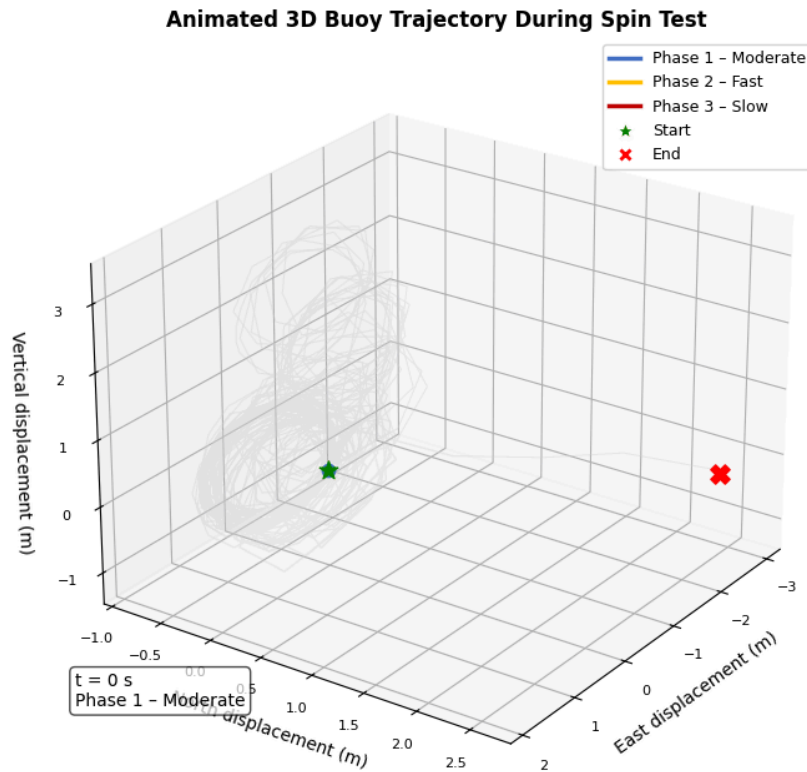


Figure 8: Animated 3D Buoy Trajectory During Spin Test

The data was parsed into the three speed phases based on our recorded timestamps of when speed was visually observed to have changed, as well as clear differences in the data's harmonics. Phase 3 recorded the most RTK Fixed data with 73.1% of points, followed by Phase 1 with 34.6%, then lastly, Phase 2 with only 3.2%. The parsed intervals shown in Figure 9 show the significance of achieving RTK Fixed for collecting clean, accurate, data. The more Fixed points there were, the more distinct a circle is plotted.

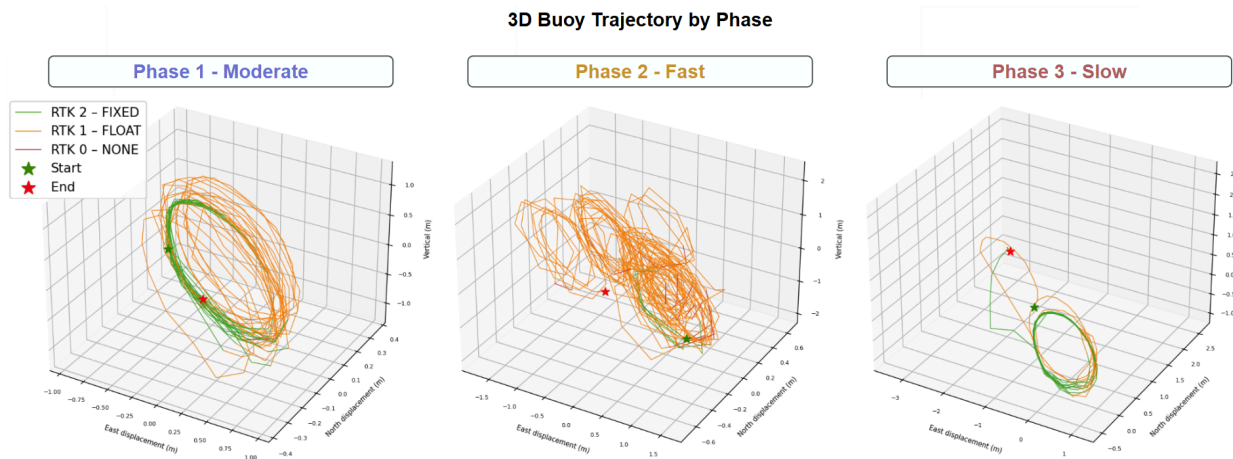


Figure 9: 3D Buoy Trajectory by Phase

Since the buoy was on a consistently rotating fixture, positional data in one coordinate can be used for further analysis. As seen in Figure 10, the buoy position along the longitudinal coordinate provides oscillatory data that can be used to find the period and frequency of the ferris wheel's motion.

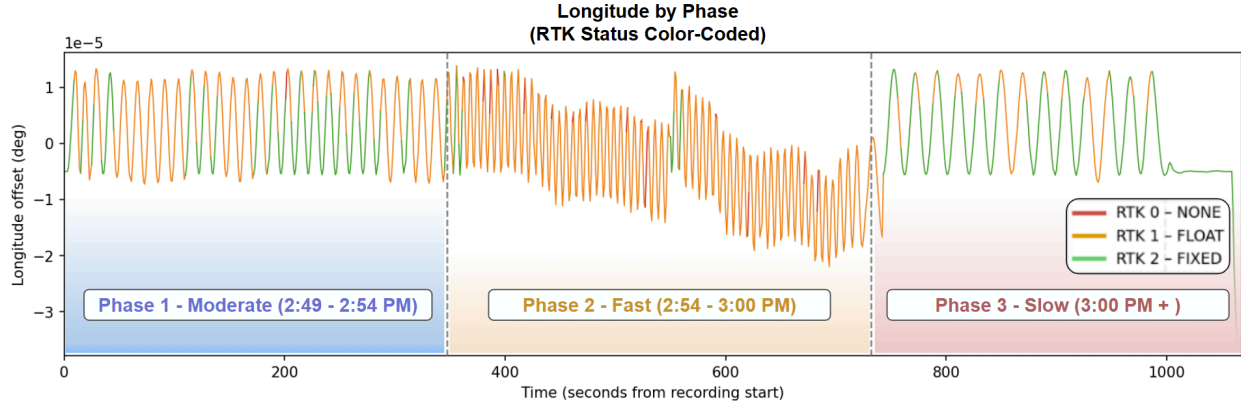


Figure 10: Longitudinal Data by Phase Intervals

With a Power Spectral Density (PSD) analysis of each individual phase, the peak frequencies indicate the frequencies with the highest energy. From these frequencies, the period was found through $T = 1/f$. For phase 1, the peak at $f = 0.084 \text{ Hz}$ corresponded to a period of $T = 11.9 \text{ s}$; for phase 2, at $f = 0.167 \text{ Hz}$, $T = 6 \text{ s}$; for phase 3, at $f = 0.055 \text{ Hz}$, $T = 18.3 \text{ s}$.

From videos taken during the experiment, the period of phase 1 was observed to be about 12 seconds, phase 2 around 6 seconds, and phase 3 around 19 seconds. This supports the data found from the PSD analysis in Figure 11.

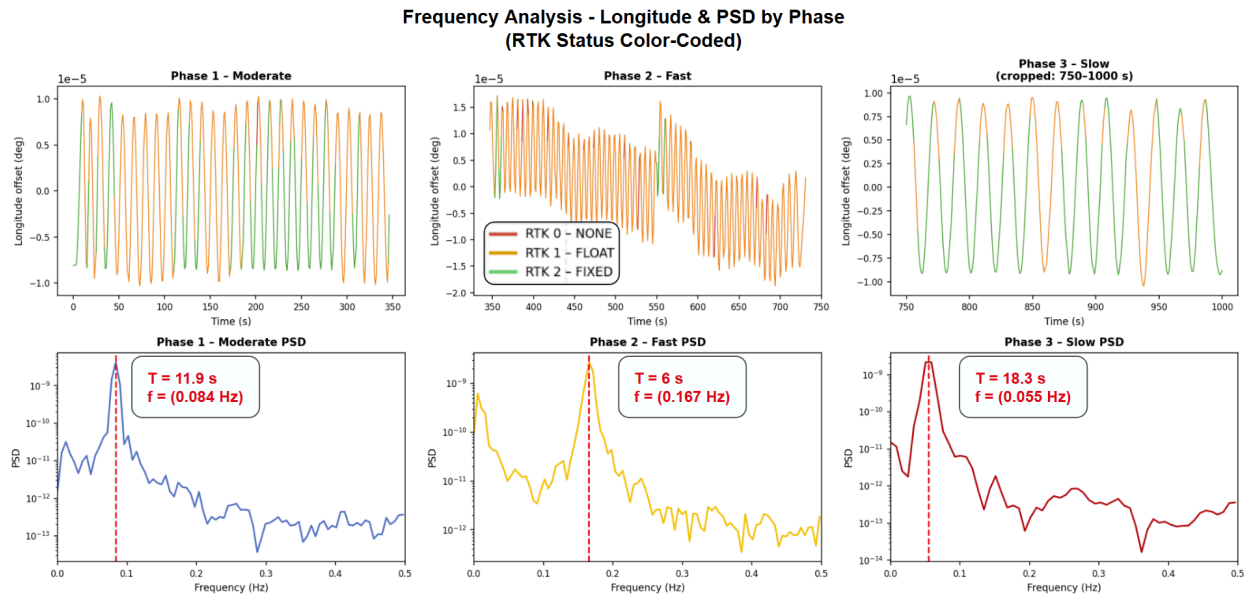


Figure 11: Power Spectral Density by Phase

Further analysis through Fast Fourier Transform (FFT) was done, converting the motion to the frequency domain to identify the dominant oscillation frequencies. The FFT amplitude spectrum of the three phases is shown in Figure 12. Similarly to the PSD analysis, the peak frequency in phase 1 was $f = 0.836 \text{ Hz}$, corresponding to a period $T = 12.0 \text{ s}$; Phase 2 had $f = 0.1653 \text{ Hz}$, with $T = 6.0 \text{ s}$; and phase 3 had $f = 0.0557 \text{ Hz}$, with $T = 17.9 \text{ s}$.

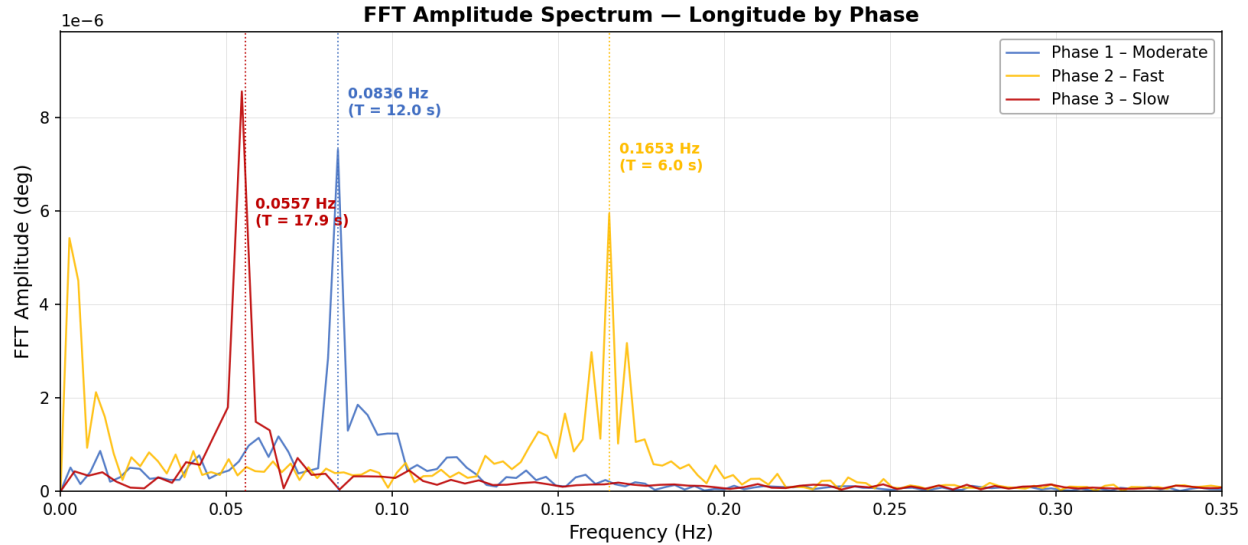


Figure 12. FFT Amplitude Spectrum by Phase

4.3 TEST 3: Salt water Pool

4.3.1. Salt water Pool Test Parameters

The saltwater pool test aimed to validate two functions: the mechanical system's reliability and the ability to record data in the fully assembled state.

For the mechanical system, the team observed how the drogue affected buoy stability. After the drogue filled with water, the buoy sat above the waterline. Dive weights (approximately 3 lbs) were added to ballast the buoy to the waterline. This maximizes stability by keeping the buoy upright and the antenna above the waterline for optimal satellite visibility. The team was also able to confirm the seal on the buoy to ensure none of the electronics were damaged and no water entered the buoy during this test.

For the electrical system, the buoy was placed in the water and subjected to jerking and pulling to verify that data logging remained uninterrupted in the fully assembled, waterborne state.

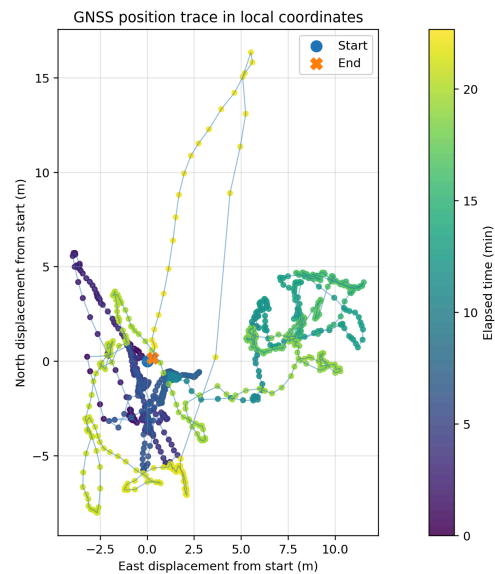
4.3.2. Salt water Pool Test Results

The mechanical system with the calibrated dive weights sat at the correct height for the buoy to resist rolling, as well as dunking when the buoy was pulled aggressively to simulate the dynamics of a rolling wave. However, the design of the internals of the buoy did not account for the integration of a cellular phone to provide wifi connectivity. As a result, the only open space in the buoy for the phone to be housed was alongside the coiled GPS antenna cables, which likely acted as a Faraday cage, preventing the phone from maintaining a consistent cellular connection.

As a result, RTK-fixed coverage was limited to roughly 40% of samples, it was not able to consistently achieve centimeter-level accuracy. Nevertheless, the position trace captured a coherent map of the buoy's relative position throughout the tug tests. Centimeter-level accuracy was demonstrated during substantial portions of the data collection. Given that the ocean deployment would have a longer duration and thus more tolerance to temporary connectivity dropouts, the system was deemed ready for field testing.

Figure 13. GNSS carrier solution accuracy as reported by the ZED-F9P receiver during the saltwater pool test.

Figure 14. Relative GPS position trace from the saltwater pool test,



showing buoy trajectory during tug tests.

4.4 TEST 4: Ocean Deployment

4.4.1. Ocean Deployment Test Parameters

The objective of this test was to validate the tracking accuracy of our low-cost custom buoy against an industry-standard benchmark: the CDIP buoy (Station 201 - Scripps Nearshore) deployed approximately 1 km offshore from Scripps Pier. Both buoys recorded concurrent time-series data of vertical ocean surface displacement (z). For this comparison, we obtained the raw GPS displacement data from CDIP, which reports east-west, north-south, and vertical displacement for each 1 Hz sample.

Our buoy was launched off of Scripps Pier from slightly above sea level and remained in the water from 3:23 PM to 4:17 PM PST, at which point it was retrieved by boat. We observed that the buoy drifted northwest, followed an arc westward, and was then swept back toward the beach (see Figure 18). The

buoy exhibited strong lateral stability and resistance to rolling, but occasionally dipped slightly underwater at the peak of larger swells.

To analyze the performance of our buoy, the raw time-series vertical displacement data from both buoys are converted into the frequency domain using a Fast Fourier Transform (FFT). This generates a wave energy density spectrum, $E(f)$, which distributes wave energy across discrete frequency bins.

The deployment is considered successful if our buoy's spectrum converges with the CDIP benchmark across three quantitative metrics:

- **Peak Period (T_p) Alignment:** The primary energy peak frequency must match, or align within a threshold of $\Delta T_p < 1.0$ second, proving the custom buoy accurately distinguishes between long-period swells and short-period wind chop.
- **Spectral Correlation (R^2):** A linear regression of the two energy density curves across the wave frequency band (0.05 to 0.5 Hz) must yield a coefficient of determination $R^2 \geq 0.85$
- **Significant Wave Height (H_s) Agreement:** The spectrally derived significant wave height, calculated from the zeroth moment of the spectrum must fall within $\pm 10\%$ of the value reported by the CDIP platform during the same deployment window.

$$H_s = 4\sqrt{\int E(f)df}$$

4.4.2. Ocean Deployment Test Results

As shown in Figure 15, the buoy maintained RTK-fixed carrier solution status throughout approximately 90% of the drift. The periods where fixed status was lost were brief (less than the Nyquist period of 1.6 s). Since these gaps are much shorter than the dominant wave period, we filled missing samples using linear interpolation between the nearest fixed-solution points on either side of each gap. Based on the way these gaps are distributed and their tendency to occur near peak altitude, we suspect it corresponds to the times our buoy briefly bobbed underwater.

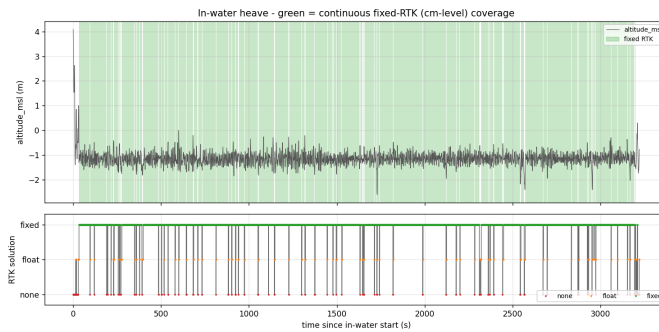


Figure 15. RTK carrier solution status over the duration of the ocean deployment. Fixed-solution coverage exceeded 90% for the in-water period.

We implemented several methods to clean and pre-process our data before computing the FFT. First, we utilized k-means clustering to identify the different segments of the buoy's deployment phases (pier, in-water, and boat retrieval) and filtered to include only the in-water segment. This method is generalizable and can be repeated for any deployment without manually selecting the analysis time period. We also trimmed 5 seconds from each end of the in-water segment to ensure the initial toss and retrieval transients do not influence the spectra. We then applied a zero-phase Butterworth bandpass filter to retain only the plausible 2–25 s wave period range and reject noise outside this band. Detrending was not applied, as it was not deemed necessary given the short monitoring duration.

We computed the FFT with a frequency resolution of 0.000397 Hz and found a dominant peak at 0.1238 Hz (period: 8.08 s) with a secondary peak at 6.3 s. To obtain a smoother spectral estimate, we also applied Welch's method, which computes the FFT over multiple overlapping segments and averages them to reduce variance. Since the short, frequent gaps in our data are not ideal for this method, we opted to linearly interpolate across the small gaps and segment the record into fewer, longer windows. The results are summarized in Figure 16 and the table below

Table 1. FFT analysis parameters

N (valid samples)	3,225
Frequency resolution	0.000397 Hz
Nyquist frequency	0.6400 Hz (min resolvable period: 1.56 s)
Dominant frequency	0.1238 Hz (period: 8.08 s)

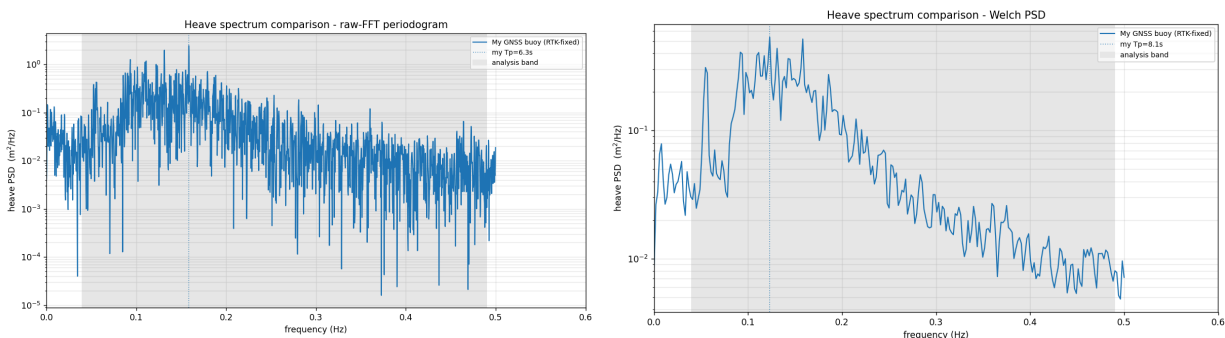


Figure 16. Wave energy density spectra computed via raw FFT (left) and Welch's method (right) for the Scripps Pier deployment (15:23–16:17 PST, June 1, 2026).

Applying the same spectral analysis to the raw CDIP displacement data over the concurrent time window, we found that all key parameters fell within the 10% accuracy threshold established in Section 4.4.1 (Table 1). For example, our peak wave period (T_p) differed from the CDIP data by less than 0.1%. The largest differences were in H_s (4.9% lower for our buoy) and peak direction. The observed differences in H_s and peak direction are consistent with the spatial separation between instruments: the CDIP buoy is

moored approximately 1 km offshore in 41 m depth, where it is insulated from nearshore refraction, shoaling, and bathymetric focusing effects that may modulate wave height and direction at the shallower deployment location of our buoy.

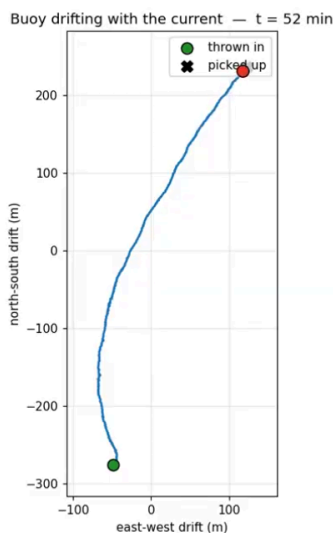


Figure 18. GPS-derived drift trajectory of the buoy during the ocean deployment.

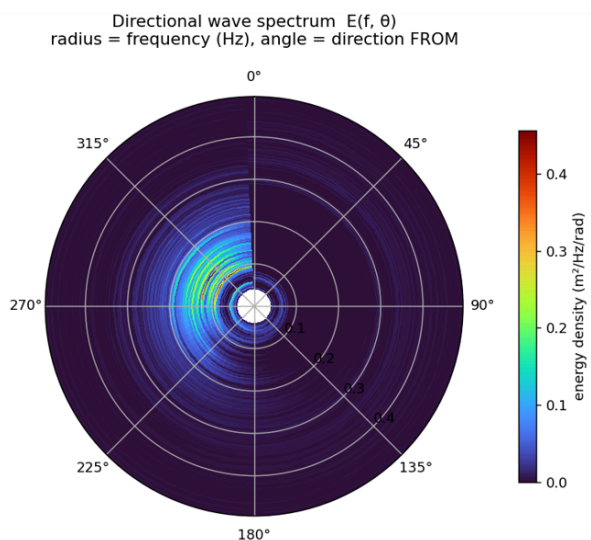


Figure 19. Directional wave spectrum showing dominant energy arriving from the northwest, consistent with the CDIP Station 201 reported peak direction of 285°.

Table 2. Spectral parameter comparison between the custom buoy and CDIP Station 201 over the concurrent deployment window.

Parameter	Our buoy	CDIP 201	diff
Hs	0.84 m	0.89 m	−4.9%
Tp (Welch)	8.1 s	8.1 s	+0.1%
Tm01 (mean period)	6.46 s	6.43 s	+0.5%
Tm02 (zero-crossing)	5.78 s	5.88 s	−1.6%
Dp (peak direction)	300°	285°	15°

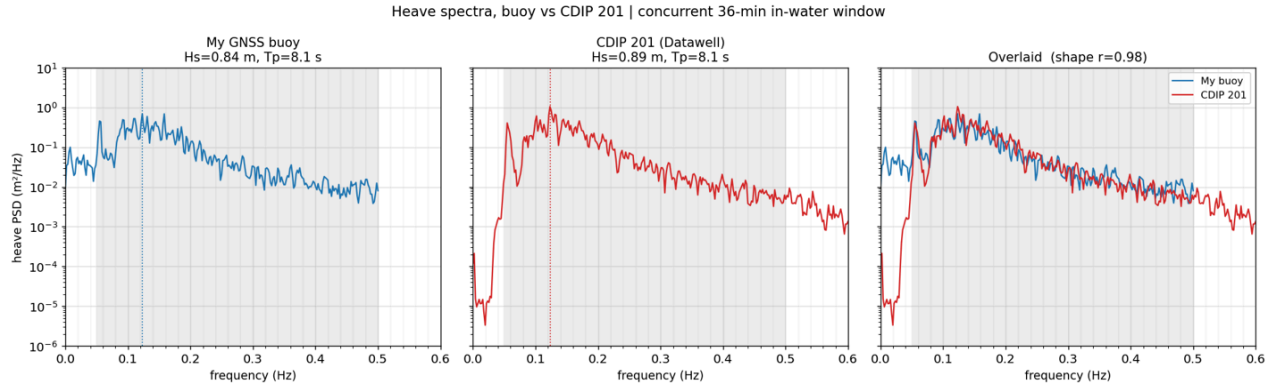


Figure 17. Comparison of MissionCreep100fan buoy (left) and CDIP buoy (center) and spectral overlay of the custom buoy and CDIP Station 201 wave energy density spectra over the concurrent 54-minute deployment window (right).

Below approximately 0.04 Hz, the two spectra diverge due to fundamental differences in how each instrument senses vertical motion. The Datawell Waverider derives heave by double-integrating vertical acceleration and applies a high-pass filter (cutoff $\approx 0.04 \text{ Hz}$) to suppress low-frequency integration noise. Our buoy measures position directly via GNSS, retaining low-frequency GPS wander and infragravity energy that the Datawell removes. This divergence lies entirely below the 0.04–0.49 Hz analysis band and therefore does not affect the computed spectral moments (H_s , T_p , T_{m01} , T_{m02}), which agree to within 5%. Within the 0.04 Hz–0.5 Hz frequency range, the spectra shows a similar distribution of frequencies, including identical peaks corresponding to an 8.1 s period.

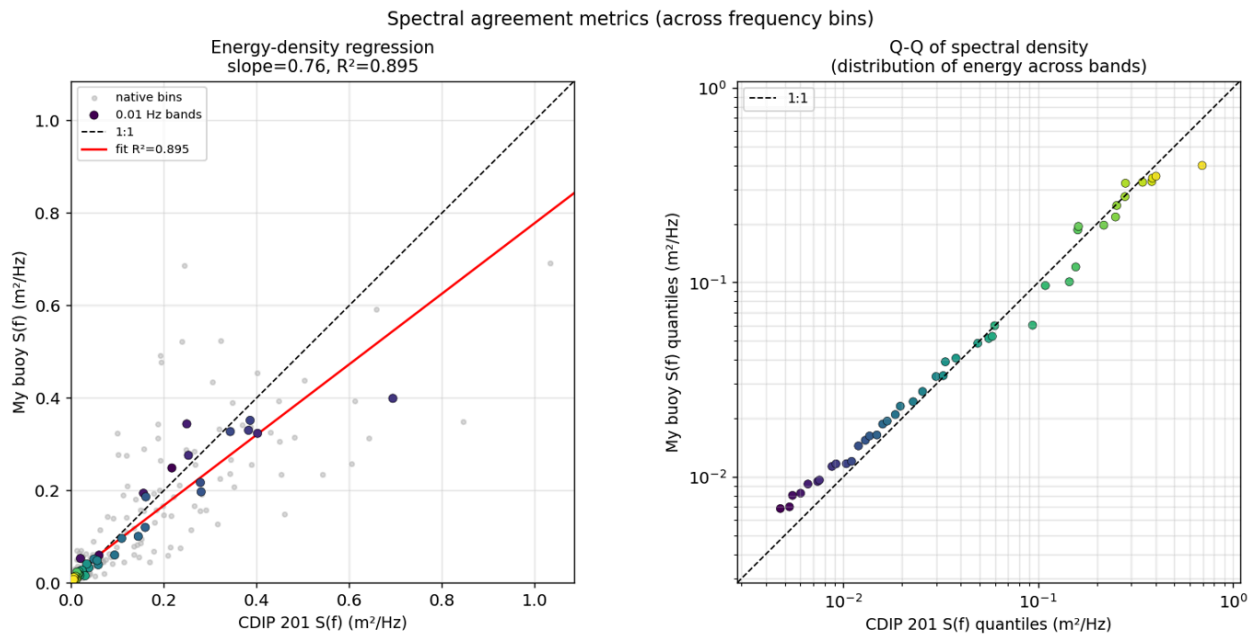


Figure 18: R -squared and Q - Q comparison of class buoy vs CDIP buoy

Across the 0.05–0.50 Hz band, regression of the two band-averaged (0.01 Hz) heave energy-density spectra gives $R^2 = 0.90$ (0.95 in log space), so 90–95% of the band-to-band variation in the buoy's spectrum is shared with the CDIP reference, meaning the two resolve the same spectral shape and peak. The fit slope of 0.76 indicates the buoy records a modestly lower energy density at the peak, consistent with its ~5% lower H_s . The quantile-quantile plot shows the two instruments report the same distribution of band energies, with points lying along the 1:1 line over nearly two decades and departing only slightly (marginally higher at the low-energy/high-frequency end and lower near the peak), as expected from the lighter GNSS buoy's greater responsiveness to short waves. Together, the high R^2 (matching spectral shape) and the near-1:1 Q-Q (matching energy magnitudes) show the buoy reproduces the reference spectrum with only a small, systematic bias rather than random error.

5. Discussion/Limitations:

Although the team was able to collect data, centimeter-level precision could not be maintained for extended periods. This is likely due to not using the backplane to mount the antenna and to the antenna cable being routed near the phone. We suspect RF interference from the cable proximity to the phone, as other teams achieved fixed-solution rates exceeding 90%. A backplane would also improve multipath rejection and signal quality. The team was also unable to eliminate the hotspot dependency, requiring a phone to be placed inside the buoy for each deployment. This limitation would be resolved by integrating an LTE modem. Future iterations could also incorporate remote data telemetry.

Mechanically, the buoy bobbed underwater and tilted, which caused noisy data. This is likely attributable to the drogue design. The drogue during saltwater pool testing showed promising self-righting behavior; however, when deployed, it behaved similarly to a sea anchor, making it slower to respond in the z direction and dragging the buoy underwater. Relocating the dive weights from the middle to the bottom of the drogue would improve self-righting behavior and reduce the tendency to submerge the buoy.

6. Conclusion:

Team MissionCreep11Fans successfully designed, built, and tested a budget-friendly, portable ocean wave buoy using off-the-shelf components. Through calibration of the components, validation of the assembly, and deployment of the buoy, the team demonstrated that a system capable of measuring wave motion using GPS and an IMU is achievable. Through this project, the team developed skills in systems integration, troubleshooting, and iterative design to meet functional requirements. The use of AI tools accelerated the data processing pipeline and helped the team overcome software challenges throughout development. In future iterations, replacing the phone hotspot with an integrated LTE modem, improving drogue stability, and strengthening the antenna ground plane would yield a more robust and accurate system.

7. Acknowledgment:

This work was performed as part of the MAE 223: Ocean Technology course taught by Professor Andrew Lucas. We thank Professor Lucas for his instruction and guidance throughout the project. We also extend our gratitude to our TA, Amara, for her valuable support, feedback, and assistance during the design, testing, and development process.

8. List of Materials and Budget

Item	Qty	Cost
Buoy Shell with Clamp Ring and O-Ring	1	~\$150.00
Li-Po Battery, 2 Cell	1	~\$40.00
Acrylic sheet 2'x4'	1	\$40.00
OpenLog Artemis	1	\$59.95
SparkFun ZED-F9P Receiver	1	\$175.00
Arduino ESP32 Thing Plus	1	\$24.95
Hardware (Misc)	N/A	~\$10.00
Ikea Bag	1	~\$1.00
Kite	1	~\$15.00
Rope (20ft spool)	1	~\$10.00

9. References

CDIP — Coastal Data Information Program. Scripps Nearshore Buoy real-time data. Scripps Institution of Oceanography, UC San Diego. <https://cdip.ucsd.edu>

SparkFun Electronics. ZED-F9P GPS-RTK2 Hookup Guide. <https://learn.sparkfun.com>

SparkFun Electronics. OpenLog Artemis Hookup Guide. <https://learn.sparkfun.com>

u-blox. ZED-F9P Product Summary — High Precision GNSS Module. u-blox AG, 2023.